

# Time and Causality from Irreversible Continuation Ordering

The Temporal and Causal Roles of Standing-Preserving Physical Persistence

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## Abstract

This paper identifies the minimal temporal and causal roles within the realized-physics arc. Previous results established that realized persistence cannot be physically null, that standing-bearing persistence localizes, that localized standing induces continuation-neighborhood domains, that stable comparison supplies the metric role of a quotient comparison carrier, that perturbation transport is bounded by continuation-comparison constraints, that fields are persistence carriers, that particles are localized persistence classes, that gauge structure is boundary-fixed presentation equivalence, that mass and inertia are deformation-response structure, and that vacuum is the minimal non-null concrete persistence baseline. The present analysis determines the ordering structure required for such a regime to continue without identity reset, record erasure, boundary-trace loss, or standing failure.

The main result is a temporal-causal role induction theorem. In a fixed admissible realized persistence regime with vacuum baseline  $\Omega_D$ , field-carrier mediation  $\Phi_D$ , particle persistence classes  $\Pi_D$ , gauge-role structure  $\mathcal{G}_D$ , mass/inertia role  $\mathcal{M}_D$ , bounded perturbation transport, and quotient comparison carrier  $C_D$ , standing-bearing irreversibility induces a temporal ordering relation  $\prec_D$ . Later standing-bearing configurations then acquire causal dependence when they require earlier fixed continuation witnesses. Thus

$$(\Omega_D, \Phi_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D, \text{BoundTrans}_D, C_D) \\ \wedge \text{IrrevCont}_D \wedge \text{RecordFix}_D \wedge \text{IdRet}_D \Rightarrow (\mathcal{T}_D, \mathcal{C}_D).$$

Physical time is the ordering role induced by irreversible standing-bearing continuation. Physical causality is the stricter dependence role by which later standing-bearing stages require earlier fixed continuation conditions. Clock time, real-valued temporal parameters, global simultaneity, Lorentzian causal cones, thermodynamic entropy laws, field equations, deterministic hidden-variable causality, and full spacetime dynamics are not assumed. They are downstream specializations of the temporal and causal roles isolated here.

## 1 Introduction

The realized-physics arc reconstructs increasingly familiar physical roles from standing-bearing persistence rather than postulating them as primitive. The first paper establishes that realized persistence cannot be physically null. The second derives bounded standing-localization as the origin of physical locality. The third derives continuation-neighborhood domains as the origin of local-domain organization. The fourth derives the metric role as quotient-stable comparative representation. The fifth derives bounded perturbation transport as the finite-propagation-constraint role. The sixth derives fields as persistence carriers. The seventh derives particles as localized persistence classes. The eighth derives gauge structure as boundary-trace-fixed

presentation equivalence. The ninth derives mass and inertia as deformation-response structure. The tenth derives vacuum as the minimal non-null concrete persistence baseline.

In compact form, the arc has established

$$\begin{aligned} \text{RealPers}_D &\Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \Rightarrow C_D \\ &\Rightarrow \text{BoundTrans}_D \Rightarrow \Phi_D \Rightarrow \Pi_D \Rightarrow \mathcal{G}_D \Rightarrow \mathcal{M}_D \Rightarrow \Omega_D. \end{aligned}$$

The next physical question is fixed by the structures already derived. Given a non-null vacuum baseline and the field, particle, gauge, mass/inertia, transport, and comparison roles, what makes continuation ordered rather than merely co-present or reversibly redescribed? What turns a retained standing consequence into a prior condition for later physical persistence? The answer is standing-bearing irreversibility, from which the temporal role is induced.

### 1.1 Theorem claim and physical role

The theorem-level claim is

$$\begin{aligned} &(\Omega_D, \Phi_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D, \text{BoundTrans}_D, C_D) \\ &\wedge \text{IrrevCont}_D \wedge \text{RecordFix}_D \wedge \text{IdRet}_D \Rightarrow (\mathcal{T}_D, \mathcal{C}_D). \end{aligned}$$

The physical-role claim is that  $\mathcal{T}_D$  is the temporal role occupied by irreversible continuation order, while  $\mathcal{C}_D$  is the causal role occupied by witness-dependent standing preservation across that order. Time is not introduced as a background line, a scalar coordinate, or a universal external parameter. Causality is not introduced as primitive influence or hidden mechanism. Both are obtained as roles internal to physically realized persistence:

*time is the ordering role induced by irreversible continuation; causality is standing-preserving dependence across that order.*

| Paper | Structural result                             | Physical role                           |
|-------|-----------------------------------------------|-----------------------------------------|
| 1     | $\text{NonVacRec}_D$                          | physical realization cannot be null     |
| 2     | $[B_D]_{\equiv_B}$                            | origin of physical locality             |
| 3     | $(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$ | origin of local-domain organization     |
| 4     | $C_D = \text{CompWit}_D / \sim_D$             | origin of metric comparison role        |
| 5     | $\text{BoundTrans}_D$                         | origin of finite-propagation constraint |
| 6     | $\Phi_D$                                      | origin of field-carrier mediation       |
| 7     | $\Pi_D = [\kappa_U]_{\equiv_\Pi}$             | origin of particle identity             |
| 8     | $\mathcal{G}_D$                               | origin of gauge structure               |
| 9     | $\mathcal{M}_D = (\mu_D, \iota_D)$            | origin of mass/inertia role             |
| 10    | $\Omega_D$                                    | origin of vacuum structure              |
| 11    | $(\mathcal{T}_D, \mathcal{C}_D)$              | origin of time and causality roles      |

### 1.2 Scope

The analysis isolates the minimal temporal and causal roles. It does not derive numerical clock time, proper time, coordinate time, Lorentzian metric time, global simultaneity, relativistic causal cones, thermodynamic entropy increase, quantum measurement collapse, deterministic causal mechanisms, field equations, causal-set dynamics, closed-timelike-curve theorems in general relativity, or global cosmic time. Those are later specializations. What is derived here is the prior physical role: irreversible continuation ordering among standing-bearing configurations, and witness-dependent causal dependence within that order.

## 2 Dependencies and imported structure

Paper 11 uses all prior layers of the realized-physics arc. The distinctive role of the present paper is to extract ordering from continuation after the regime already has field-carrier mediation, particle identity, gauge-compatible presentation, mass/inertia response, and vacuum baseline structure. The dependency order is summarized in Table 1.

Table 1: Imported structure for the temporal-causal derivation.

| Imported result                      | Role in the present paper                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| Non-vacuous realized persistence     | Time cannot order nullity; ordering requires standing-bearing physical realization.                   |
| Bounded standing-localization        | Temporal and causal roles respect bounded local persistence rather than global undifferentiated flow. |
| Continuation-neighborhood domains    | Ordering occurs over local continuation domains and their admissible relations.                       |
| Quotient comparison carrier $C_D$    | Comparison of stages and retained consequences is quotient-stable, not primitive coordinate time.     |
| Bounded perturbation transport       | Causal dependence inherits bounded transport and no pathless global effect.                           |
| Field-carrier mediation $\Phi_D$     | Continuation stages require carrier-mediated persistence, not bare descriptions.                      |
| Particle persistence classes $\Pi_D$ | Particle histories require identity retention across ordered continuation.                            |
| Gauge-role structure $\mathcal{G}_D$ | Temporal and causal claims must be gauge-presentation invariant.                                      |
| Mass/inertia role $\mathcal{M}_D$    | Deformation and transport-state response persist across ordered change.                               |
| Vacuum baseline $\Omega_D$           | No-particle continuation stages remain non-null physical stages.                                      |
| Persistence-regime realization       | Supplies irreversible continuation participation and consequence retention.                           |
| Admissible record construction       | Supplies record fixation as downstream physical witness, not observer-memory primitive.               |
| Bivalent trajectory theory           | Supplies standing-preserving trajectory order and fail-closed continuation.                           |
| Boundary trace fixation              | Prior-stage availability must be boundary-fixed, not label-fixed.                                     |
| AMetric boundary                     | Blocks primitive time, global simultaneity selectors, and metric-time import before ordering.         |

The key methodological point is that temporal and causal roles are not imported from standard spacetime formalisms. Relativistic spacetime, causal curves, causal sets, thermodynamic arrows, and clock parameters are later mathematical specializations of ordering and dependence roles. They are not assumed in the derivation.

### 3 Definitions and type discipline

**Definition 3.1** (Continuation stage). A continuation stage is a standing-bearing role configuration

$$e_i = (U_i, \Phi_i, \Pi_i, \mathcal{G}_i, \mathcal{M}_i, \Omega_i, \beta_i, r_i),$$

where each component is inherited from a prior structural layer of the arc. A stage is not an instantaneous spacetime slice; it is the role configuration whose standing-bearing continuation is evaluated.

| Stage component | Prior role forcing the component                     |
|-----------------|------------------------------------------------------|
| $U_i$           | continuation-neighborhood and local-domain structure |
| $\Phi_i$        | field-carrier mediation                              |
| $\Pi_i$         | particle persistence-class identity                  |
| $\mathcal{G}_i$ | boundary-fixed gauge-role structure                  |
| $\mathcal{M}_i$ | mass/inertia deformation-response profile            |
| $\Omega_i$      | non-null vacuum baseline                             |
| $\beta_i$       | bounded perturbation transport/restoration           |
| $r_i$           | record or boundary-trace fixation                    |

**Definition 3.2** (Standing-bearing continuation). A transition

$$e_i \rightarrow_D e_j$$

is standing-bearing iff it preserves admissible identity, boundary trace, transport/restoration profile, field-carrier mediation, and the relevant particle, vacuum, gauge, and mass/inertia roles. Standing-bearing continuation is persistence of physically operative role structure, not mere succession of labels.

**Definition 3.3** (Irreversible continuation). For standing-bearing stages  $e_i, e_j$ , write

$$e_i \prec_D e_j$$

iff  $e_j$  admissibly continues  $e_i$ , retains the standing-bearing consequence of  $e_i$ , and removal or reversal of  $e_i$ 's contribution would require identity reset, record erasure, boundary-trace loss, gauge-role break, transport incompatibility, or standing failure. This is a persistence condition first: it says that the prior contribution is non-dischargeable inside the realized continuation regime. The temporal role is induced by this non-dischargeability; it is not assumed in advance.

**Definition 3.4** (Temporal role). The temporal role of a regime is

$$\mathcal{T}_D = (E_D, \prec_D),$$

where  $E_D$  is the family of standing-bearing continuation stages and  $\prec_D$  is irreversible continuation order. At this level  $\prec_D$  is partial, local, standing-relative, and non-metric. It is not a clock parameter and need not be a total order. Two stages may remain unordered unless an admitted continuation path, transport witness, record fixation, or boundary trace relates them inside the fixed regime.

**Definition 3.5** (Causal dependence). For standing-bearing stages  $e_i, e_j$ , write

$$e_i \rightsquigarrow_D e_j$$

iff (i)  $e_i \prec_D e_j$ , (ii)  $e_j$ 's standing-bearing status requires a fixed witness from  $e_i$ , and (iii) removing  $e_i$  changes the admissible continuation class of  $e_j$ . The causal role is

$$\mathcal{C}_D = (E_D, \rightsquigarrow_D).$$

Causal dependence is stricter than temporal priority:

$$e_i \rightsquigarrow_D e_j \Rightarrow e_i \prec_D e_j,$$

but not every temporally prior stage is a cause of every later stage. Physical causality first enters here as admissible standing-bearing dependence; field-mediated, relativistic, dynamical, or propagator-based causality are later specializations.

**Definition 3.6** (Record fixation). A record is a boundary-fixed retained continuation witness available for admissible downstream reuse. Record fixation is not primitive time, observer memory, semantic representation, or measurement-only description. It is the admissible physical availability of a prior standing-bearing contribution for later continuation.

## 4 Irreversibility before temporal representation

The derivation begins with a persistence condition: a standing-bearing contribution is retained in another admissible configuration in such a way that its removal would alter identity, boundary fixation, transport compatibility, or standing status. Temporal order is the physical role induced by this pre-temporal persistence asymmetry.

| Persistence irreversibility                 | Temporal role induced       |
|---------------------------------------------|-----------------------------|
| retained consequence                        | earlier/later ordering role |
| standing-preserving continuation asymmetry  | temporal relation           |
| boundary/record fixation                    | prior-stage availability    |
| non-removability without standing failure   | temporal asymmetry          |
| identity-preserving continuation constraint | ordered persistence history |

**Proposition 4.1** (Irreversibility-first grounding). *Irreversibility is defined without assuming primitive temporal order. It is fixed by standing-bearing consequence retention, boundary-fixed record availability, and the inability to remove a prior contribution without identity reset, boundary-trace loss, transport incompatibility, or standing failure. Temporal order is the ordering role induced by that persistence asymmetry.*

*Proof.* The defining conditions for  $\prec_D$  concern persistence roles: retained consequence, record availability, boundary trace, transport compatibility, and identity preservation. None requires a clock parameter, global simultaneity relation, metric time coordinate, or external temporal line. Once such non-dischargeable consequence is present, one contribution occupies the prior role relative to another because the later standing-bearing stage depends on the retained contribution. Thus irreversibility is the persistence condition, and temporal order is the role it induces.  $\square$

## 5 Failure of primitive temporal alternatives

**Theorem 5.1** (No primitive temporal parameter assignment). *A temporal parameter assignment  $t$  introduced before irreversible continuation ordering is fixed functions only as a selector or coordinate skin. If  $t$  represents  $\prec_D$ , it is a later representation of the temporal role. If it does not represent  $\prec_D$ , it carries no temporal role inside the realized persistence regime.*

*Proof.* A primitive temporal assignment can order labels without establishing which standing-bearing consequences are retained, which records are fixed, which boundary traces remain available, or which later stages depend on earlier ones. Such ordering is presentational unless tied to irreversible continuation. If the assignment is tied to  $\prec_D$ , then the physical temporal role is supplied by  $\prec_D$  and the assignment is representational. If it is not tied to  $\prec_D$ , it cannot distinguish retained physical continuation from arbitrary indexing. Hence primitive temporal-parameter assignment fails as a source of temporal role.  $\square$

**Lemma 5.2** (Reversible continuation collapse). *If standing-bearing continuation is fully reversible without residue, no continuation stage is fixed as a non-removable contribution to another standing-bearing configuration. Record fixation, identity-memory, and retained persistence consequence therefore collapse.*

*Proof.* Suppose every continuation contribution can be reversed without residue while preserving the same standing-bearing status. Then no stage retains a non-dischargeable contribution into another standing-bearing configuration. Boundary traces can be erased without loss, records carry no downstream constraint, and identity-memory is indistinguishable from reversible co-description. The temporal role requires precisely the contrary: retained consequence that cannot be removed without changing the admissible standing of the continuing configuration. Thus fully reversible residue-free continuation cannot supply the temporal role.  $\square$

**Lemma 5.3** (Default simultaneity collapse). *If all standing-bearing stages are treated as co-present without irreversible order, perturbation transport, record fixation, restoration, and identity-memory cannot distinguish retained consequence from arbitrary co-description.*

*Proof.* Default co-presence removes the relation by which one stage becomes fixed relative to another. Without such ordering, a restoration cannot require a fixed antecedent witness, a transported perturbation cannot be inherited as retained burden, and an identity-memory profile cannot distinguish standing-bearing continuation from simultaneous presentation. The result is either a global selector, a block description without internal continuation role, or standing failure. Therefore simultaneity-by-default cannot ground the temporal role of a physically realized persistence regime.  $\square$

## 6 Temporal order from irreversible continuation

**Proposition 6.1** (Temporal ordering from irreversible continuation). *If  $e_i \prec_D e_j$ , then  $e_i$  is temporally prior to  $e_j$  in the minimal physical sense:  $e_j$  retains standing-bearing consequence of  $e_i$ , and  $e_i$ 's contribution cannot be removed without changing  $e_j$ 's admissible status.*

*Proof.* By definition,  $e_i \prec_D e_j$  requires retained consequence, admissible continuation, and non-removability without identity reset, record erasure, boundary-trace loss, transport incompatibility, or standing failure. Proposition 4.1 establishes that these conditions are persistence conditions

rather than hidden clock assumptions. They supply the ordering role required for temporal priority before clock parametrization. Hence irreversible continuation induces temporal ordering.  $\square$

**Proposition 6.2** (Local partiality of temporal order). *The temporal relation  $\prec_D$  need not be total. Two stages may remain unordered unless a continuation path, transport witness, record fixation, or boundary trace relates them inside the fixed regime.*

*Proof.* A total order would compare every pair of stages whether or not the regime supplies an admissible continuation relation between them. Such comparison would import a global time selector. The prior arc already excludes pathless transport, hidden comparison carriers, and default neighborhood relations. Therefore  $\prec_D$  is induced only where standing-bearing irreversible continuation is witnessed. Its natural form at this level is local and partial.  $\square$

**Proposition 6.3** (Record-fixed retention). *If  $e_j$  uses  $e_i$  as a fixed standing-bearing witness, then  $e_i$ 's contribution is retained as a fixed continuation contribution in  $e_j$ .*

*Proof.* A fixed witness is not merely an annotation. It is a boundary-fixed downstream availability condition. If  $e_j$ 's standing status depends on that witness, erasing or replacing  $e_i$ 's contribution changes the admissible continuation class of  $e_j$ . The antecedent contribution is therefore retained as part of  $e_j$ 's standing-bearing continuation.  $\square$

## 7 Causal dependence as stricter-than-time ordering

Temporal priority alone does not imply causation. A stage can be prior without being required by a later stage. Causal role begins when later standing-bearing status depends on an earlier fixed continuation witness.

**Proposition 7.1** (Causal dependence requires temporal order). *If  $e_i \rightsquigarrow_D e_j$ , then  $e_i \prec_D e_j$ .*

*Proof.* By definition, causal dependence requires that  $e_j$ 's standing-bearing status require a fixed witness from  $e_i$ . Such witness dependence is meaningful only if  $e_i$ 's contribution is retained as prior relative to  $e_j$ . Hence causal dependence entails temporal priority. The converse does not follow: a prior stage may be historically retained without being required for a given later standing status.  $\square$

**Theorem 7.2** (Causal role induction). *In a fixed admissible realized persistence regime, if a later standing-bearing stage requires an earlier fixed continuation witness, causal dependence is induced:*

$$(e_i \prec_D e_j) \wedge \text{WitnessReq}_D(e_i, e_j) \Rightarrow e_i \rightsquigarrow_D e_j.$$

*Proof.* The witness requirement states that removing or altering the earlier fixed contribution changes the admissible standing class of the later stage. Since  $e_i \prec_D e_j$ , the contribution is already temporally prior by irreversible continuation. The additional requirement that  $e_j$ 's standing depends on the fixed witness supplies the causal role. Thus causal dependence is not primitive influence; it is witness-dependent standing preservation across irreversible continuation.  $\square$

## 8 Temporal-causal role induction

**Theorem 8.1** (Temporal-causal role induction). *In a fixed admissible realized persistence regime with vacuum baseline, field-carrier mediation, particle persistence classes, gauge role, mass/inertia role, bounded perturbation transport, and quotient comparison carrier, standing-bearing irreversible continuation induces temporal order, and witness-dependent continuation induces causal order:*

$$(\Omega_D, \Phi_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D, \text{BoundTrans}_D, C_D) \\ \wedge \text{IrrevCont}_D \wedge \text{RecordFix}_D \wedge \text{IdRet}_D \Rightarrow (\mathcal{T}_D, \mathcal{C}_D).$$

*Proof.* The prior arc supplies non-null physical persistence, bounded localization, local continuation domains, quotient comparison, bounded transport, field-carrier mediation, particle identity, gauge-compatible presentation, mass/inertia response, and vacuum baseline structure. Standing-bearing continuation of these structures requires retained consequence. If retained consequence can be removed or reversed without residue, Lemma 5.2 applies and no temporal role is induced. Under  $\text{IrrevCont}_D$ , however, retained consequence supplies  $\prec_D$ , the irreversible continuation relation. Record and boundary fixation make prior-stage contributions available as fixed witnesses for later continuation. When a later stage requires such a fixed witness, Theorem 7.2 induces  $\rightsquigarrow_D$ . The resulting structures  $\mathcal{T}_D = (E_D, \prec_D)$  and  $\mathcal{C}_D = (E_D, \rightsquigarrow_D)$  are the temporal and causal roles of the regime.  $\square$

**Proposition 8.2** (No pathless causal dependence). *A causal relation cannot be assigned between  $e_i$  and  $e_j$  without an admitted continuation, transport, boundary, or record witness. Otherwise the alleged dependence is hidden selector import, unsupported global causal assignment, or cross-domain influence.*

*Proof.* Causal dependence requires witness-dependent standing preservation. If no continuation, transport, boundary, or record witness connects the stages, the dependence cannot be fixed inside the regime. Assigning it anyway would introduce a hidden selector or pathless causal edge. This violates bounded perturbation transport and boundary fixation. Hence causal dependence is admissible only where the relevant witness structure exists.  $\square$

**Corollary 8.3** (No closed reversal of standing consequence). *A closed loop*

$$e_0 \prec_D e_1 \prec_D \cdots \prec_D e_0$$

*is inadmissible if it requires erasing or undoing fixed standing-bearing consequence.*

*Proof.* Each strict relation  $\prec_D$  encodes retained consequence that cannot be removed without changing later admissible standing. Returning to  $e_0$  while erasing the fixed consequence of the cycle would contradict irreversibility. If a loop is merely a representational cycle with no standing consequence erased, it is not a closed reversal of  $\prec_D$ . This corollary is a fixed-domain standing consequence result, not a theorem about general-relativistic closed timelike curves.  $\square$

## 9 Relations to standard temporal and causal structures

### 9.1 Thermodynamic arrow

The thermodynamic arrow is not assumed. Entropy increase, coarse-graining, state-counting, and energetic bookkeeping require additional structure not derived here. The present result supplies

the prior role that thermodynamic arrows later specialize: irreversible continuation asymmetry, retained consequence, record-fixation asymmetry, and persistence-history ordering. Statistical and energetic formalisms can quantify that role only after the continuation order is already available.

## 9.2 Relativistic causality

The paper does not derive Lorentzian causal cones, proper time, spacetime intervals, global hyperbolicity, or relativistic causal order. It supplies the prior roles that such structures must represent: local irreversible ordering and bounded witness-mediated dependence. In spacetime physics, causal curves and light cones are metric-geometric specializations of ordering and dependence roles.

## 9.3 Quantum measurement and records

Record fixation is not measurement collapse. An admissible record is a boundary-fixed continuation witness. Detector records and measurement outcomes may specialize record fixation, but observer memory and measurement reports are not primitive sources of time. This distinction is important in nonclassical settings where correlations and boundary compatibility need not reduce to Bell-local hidden variables.

## 9.4 Static descriptions and block representations

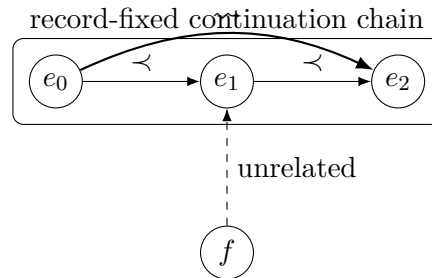
The result does not deny static mathematical representations. A block description may represent an ordered history. It cannot replace the physical role of irreversible continuation if the regime itself contains standing-bearing persistence. The internal role is not the existence of a diagram with all stages drawn at once; it is the retained consequence relation by which later stages stand on earlier fixed continuation conditions.

# 10 Finite witness model

A finite witness model makes the distinction between temporal order and causal dependence explicit. Let

$$H_D = (E, \prec, \rightsquigarrow, r, \tau)$$

where  $E$  is a finite set of standing-bearing stages,  $\prec$  is irreversible continuation order,  $\rightsquigarrow$  is witness-dependent causal relation,  $r$  is record/boundary fixation, and  $\tau$  is admitted transport witness structure.



The model has the following collapse cases.

1. No  $\prec$ : there is no temporal role.

2. Reversible loop: fixed consequence is erased, so standing-bearing irreversibility fails.
3. Total co-presence: retained history is indistinguishable from co-description.
4. Causal edge without witness: selector import or pathless influence.
5. Temporal order without witness dependence: local temporal ordering is present, but causal dependence does not follow.
6. Witness-dependent continuation: causal role is induced.

The witness graph instantiates one admissible finite realization of temporal-causal ordering; it does not define time or causality as graph data.

## 11 Route exhaustion

Any proposed source of time and causality inside a realized persistence regime is one of the following:

1. irreversible continuation ordering;
2. primitive temporal-parameter assignment;
3. reversible co-description;
4. simultaneity-by-default;
5. observer-record chronology;
6. pathless causal influence;
7. metric-time import before ordering.

Options 2–7 fail as primitive sources. A temporal parameter is coordinate skin unless it represents irreversible order. Reversible co-description and simultaneity-by-default cannot ground retained consequence. Observer-record chronology mistakes empirical or semantic records for the physical source of ordering. Pathless influence violates bounded transport and boundary fixation. Metric-time import presupposes the ordering role it is supposed to explain. The surviving source is irreversible standing-bearing continuation.

## 12 Normal form

The result has the following normal form:

$$\boxed{(\Omega_D, \Phi_D, \Pi_D, \mathcal{G}_D, \mathcal{M}_D, \text{BoundTrans}_D, C_D) \wedge \text{IrrevCont}_D \wedge \text{RecordFix}_D \wedge \text{IdRet}_D \Rightarrow (\mathcal{T}_D, \mathcal{C}_D)}$$

with

$$\mathcal{T}_D = (E_D, \prec_D), \quad \mathcal{C}_D = (E_D, \rightsquigarrow_D),$$

and

$$e_i \rightsquigarrow_D e_j \Rightarrow e_i \prec_D e_j.$$

Temporal order is irreversible continuation order. Causal dependence is the stricter witness-dependent relation across that order.

## 13 Conclusion

Paper 1 established that realized persistence cannot be physically null. Paper 2 established bounded localization. Paper 3 derived continuation-neighborhood domains. Paper 4 derived the metric role of quotient-stable comparison. Paper 5 derived bounded perturbation transport. Paper 6 derived fields as persistence carriers. Paper 7 derived particles as localized persistence classes. Paper 8 derived gauge structure as boundary-fixed presentation equivalence. Paper 9 derived mass and inertia as deformation-response structure. Paper 10 derived vacuum as the minimal concrete persistence baseline.

The present analysis isolates the minimal irreversible ordering and witness-dependent dependence structure required for standing-bearing physical continuation. Physical time is the ordering role induced by irreversible continuation. Physical causality is the stricter dependence role by which later standing-bearing stages require earlier fixed continuation witnesses. Clock time, relativistic causal cones, thermodynamic arrows, field equations, and global spacetime time structures are downstream specializations of these roles. Time and causality first enter realized physics as irreversible ordering and witness-dependent continuation among standing-bearing physical configurations.

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